

Problem Set 1

CMSC 27200: Theory of Algorithms

Assigned January 8, 2025, Due January 17, 2025

Problem 1: Gale-Shapley and Stable Matchings

- (a) Given the following preference lists, run the Gale-Shapley algorithm with group A making the offers to obtain a stable matching.

Group A 's preference lists (from most preferred to least preferred):

a_1 : b_3, b_2, b_4, b_1

a_2 : b_4, b_1, b_3, b_2

a_3 : b_4, b_1, b_2, b_3

a_4 : b_1, b_4, b_3, b_2

Group B 's preference lists (from most preferred to least preferred):

b_1 : a_1, a_3, a_4, a_2

b_2 : a_3, a_4, a_2, a_1

b_3 : a_4, a_2, a_3, a_1

b_4 : a_1, a_4, a_3, a_2

For your answer (and for your answer to part b as well), you should give the following data. At each step, give the offer that is made, whether or not this offer is accepted, and the resulting partial matching.

For this part (and for part b), you may write a computer program to implement the Gale-Shapley algorithm and generate the needed data. If you do so, you should submit your code along with the data.

- (b) Now run the Gale-Shapley algorithm with group B making the offers to obtain another stable matching. Which people are happier in this new stable matching (compared to the stable matching found in part a)?
- (c) **Extra credit (a correct answer increases your score by 1, there is no penalty for an incorrect or incomplete answer):** What other stable matching(s) are there, if any? Note: To earn this extra credit, you should show that you have indeed found all of the possible stable matchings. Hint is available.

Problem 2: Big O Notation

- (a) Given each of the following statements, what can we say about $T(n)$ using Big O notation? You are asked to give the best possible answers using the concepts of $O(\cdot)$, $\Omega(\cdot)$, and/or $\Theta(\cdot)$. All logarithms are base 2.

1. $T(n) \leq n^{1.5} + 2n^2 + 3$.
2. $T(n) = 2\sqrt{n} + 10(\log n)^5$.
3. $T(n) \geq 10n^4 + 3n(1.1)^n$.
4. $2^n \leq T(n) \leq 7n^2 2^{2n}$.
5.
$$T(n) = \begin{cases} 14 - 5n + \frac{n^2}{2} & \text{if } n \bmod 2 = 0 \\ 3n \log n + 100n & \text{if } n \bmod 2 = 1 \end{cases}$$

- (b) Order the functions below in order of increasing asymptotic growth. If two functions have the same asymptotic growth, you should indicate this. All logarithms are base 2.

3^n , n , $2 \log n$, $(\log n)^{\log n}$, n^2 , $6n \log n$, $5n^{\log \log n}$, $2^{2 \log n}$, $n2^n$, $2^{\log \log n}$, $2^{\sqrt{\log n}}$
Hint is available.

Problem 3: A New Partner in Town

In this problem, we consider the stable matching problem when the two groups of people $A = \{a_1, \dots, a_m\}$ and $B = \{b_1, \dots, b_n\}$ may have different sizes (i.e., m may not be equal to n).

To do this, we generalize the definition of an unstable pair.

Definition 0.1. Given a matching M between A and B , we define an unstable pair to be a pair of people (a_i, b_j) such that

1. a_i is not matched with b_j .
2. a_i is either unmatched or prefers b_j to his/her current partner.
3. b_j is either unmatched or prefers a_i to his/her current partner.

We say that a matching M is stable if there are no unstable pairs.

Remark 0.2. It is not hard to show that the Gale-Shapley algorithm can also be used to find a stable matching when A and B have different sizes. One way to show this is to add dummy people to the smaller group who are at the end of the preference lists of everyone in the larger group (the preferences of the dummy people are arbitrary). We can then run Gale-Shapley as before where anyone who ends up with a dummy person is instead unmatched at the end. If the resulting matching had an unstable pair, this would also be an unstable pair when the dummy people are included which contradicts the fact that Gale-Shapley always gives a stable matching.

For each of the following statements, say whether the statement is true or false. If the statement is true, explain why it is true. If the statement is false, give a counterexample.

- (a) Let $A' = \{a_1, \dots, a_n, a_{n+1}\}$ be a group of $n + 1$ people and let $B = \{b_1, \dots, b_n\}$ be a group of n people. If a_i is the person in group A' who is unmatched when we run Gale-Shapley with group A' making the offers then there is no stable matching between A' and B where a_i has a partner.
- (b) Let $A = \{a_1, \dots, a_n\}$, $B = \{b_1, \dots, b_n\}$, and let $M = \{(a_i, b_{\pi(i)}) : i \in [n]\}$ be the matching found by Gale-Shapley for A and B with group A making the offers (where $\pi : [n] \rightarrow [n]$ is a permutation).

Let's say that a new person a_{n+1} is added to group A , resulting in a larger group $A' = \{a_1, \dots, a_n, a_{n+1}\}$ (a_{n+1} can be inserted anywhere in the preference lists for group B). If we run Gale-Shapley on groups A' and B and let M' be the resulting matching then everyone in group B is at least as happy with M' as they would have been with M .

More precisely, write $M' = \{(a_i, b_{\pi'(i)}) : i \in [n+1] \setminus i_{\text{leftover}}\}$ where $i_{\text{leftover}} \in [n+1]$ is the index such that $a_{i_{\text{leftover}}}$ is unmatched and $\pi' : [n+1] \setminus i_{\text{leftover}} \rightarrow [n]$ is a bijection. For all $j \in [n]$, either $\pi'^{-1}(j) = \pi^{-1}(j)$, or b_j prefers $a_{\pi'^{-1}(j)}$ to $a_{\pi^{-1}(j)}$.

- (c) As in part b, let $A = \{a_1, \dots, a_n\}$, $B = \{b_1, \dots, b_n\}$, and let $M = \{(a_i, b_{\pi(i)}) : i \in [n]\}$ be the matching found by Gale-Shapley for A and B with group A making the offers (where $\pi : [n] \rightarrow [n]$ is a permutation).

Let's say that a new person b_{n+1} is added to group B , resulting in a larger group $B' = \{b_1, \dots, b_n, b_{n+1}\}$ (b_{n+1} can be inserted anywhere in the preference lists for group A). If we run Gale-Shapley on groups A and B' and let M' be the resulting matching then everyone in group A is at least as happy with M' than they would have been with M .

More precisely, write $M' = \{(a_i, b_{\pi'(i)}) : i \in [n]\}$ where $\pi' : [n] \rightarrow [n+1]$ is injective. For all $i \in [n]$, either $\pi'(i) = \pi(i)$ or a_i prefers $b_{\pi'(i)}$ to $b_{\pi(i)}$.

[Note: It is possible to earn an exemplary grade on one part of this problem.]

A hint is available for this problem.

Problem 4: Stable Matching on a Line

In this problem, we consider the special case of the stable matching problem where people's preferences are completely determined by their position on a line. The setup is as follows:

1. As before, $A = \{a_1, \dots, a_n\}$ and $B = \{b_1, \dots, b_n\}$ are two groups of n people.
2. Each person $a_i \in A$ has a position $x_i \in \mathbb{R}$. The people in group A are labeled so that $x_1 < x_2 < \dots < x_n$.
3. Each person $b_j \in B$ has a position $y_j \in \mathbb{R}$. The people in group B are labeled so that $y_1 < y_2 < \dots < y_n$.
4. A person $a_i \in A$ prefers b_j over $b_{j'}$ if he/she is closer to b_j than $b_{j'}$ (i.e., $|x_i - y_j| < |x_i - y_{j'}|$). We assume that we never have any ties (i.e., there are no $i, j, j' \in [n]$ such that $j' \neq j$ and $|x_i - y_j| = |x_i - y_{j'}|$).

5. A person $b_j \in B$ prefers a_i over $a_{i'}$ if he/she is closer to a_i than $a_{i'}$ (i.e., $|y_j - x_i| < |y_j - x_{i'}|$). We assume that we never have any ties (i.e., there are no $i, i', j \in [n]$ such that $i' \neq i$ and $|y_j - x_i| = |y_j - x_{i'}|$).

A potential greedy algorithm for this problem is to take the matching $\{(x_i, y_i) : i \in [n]\}$. In this problem, we consider whether this is a good matching or not.

- (a) Given a matching $M = \{(a_i, b_{\pi(i)}) : i \in [n]\}$, we define the total distance of M to be $D = \sum_{i=1}^n |x_i - y_{\pi(i)}|$. Explain why/prove that the total distance of the matching $\{(x_i, y_i) : i \in [n]\}$ is less than or equal to the total distance of any other matching $M = \{(a_i, b_{\pi(i)}) : i \in [n]\}$.

[Note: It is possible to earn an exemplary grade on this part of the problem.]

Hint is available.

- (b) Is the matching $\{(x_i, y_i) : i \in [n]\}$ always stable? If so, explain why. If not, give a counterexample.
- (c) **Extra credit (a correct answer increases your score by 1, there is no penalty for an incorrect or incomplete answer):** In this setting, is there always a unique stable matching? If so, explain what the unique stable matching is and why there are no other stable matchings. If not, give a counterexample.

Acknowledgment: This problem is similar to a problem from Andy Drucker.

Hints:

1c. Use parts a and b and the properties of the Gale-Shapley algorithm to determine the set of possible partners for each person in a stable matching and then work from there.

2b. Taking the logarithm of these functions may be helpful for comparing them.

3. The following properties of the Gale-Shapley algorithm may be helpful:

1. As Gale-Shapley progresses, the people in group B can only become happier.
2. As Gale-Shapley progresses, for each person $a_i \in A$, the set of people $b_j \in B$ who are available to a_i can only shrink.
3. As long as group A makes the offers and the people in group A make the offers based on their preference lists, the final matching (and the set of offers which were made) is independent of the order in which the offers were made.
4. The stable matching given by Gale-Shapley is the best possible stable matching for everyone in group A and the worst possible stable matching for everyone in group B .

4a. One way to prove this is by using an exchange argument.